Frontier



Submarine Scientific

R&D application

2026

General Application

The General Application applies to everyone; all applicants should complete this section

Public section

The content in this section (answers to questions 1(a) - (e)) will be made public on the Frontier GitHub repository after we announce funding decisions in the summer and fall. Include as much detail as possible but omit sensitive and proprietary information.

Organization name

Submarine Scientific

Organization location (we welcome applicants from anywhere in the world)

San Francisco, California, USA

Name(s) of primary point(s) of contact for this application

Veronica Tamsitt, Jonathan Kirk, Jacki Long

Brief organization description, including organization type (e.g., for-profit, NGO, research institution/university) <20 words

Submarine is a for-profit organization building the scientific foundation that mCDR needs to earn trust and scale responsibly

1. Public summary of proposed project[1] to Frontier

a. **Project title** <20 words

Unlocking GPU-native regional ocean modeling for ocean-based carbon removal

[1] We use “project” throughout this template, but the term is not intended to denote a single facility or experimental setup. The “project” being proposed to Frontier could include multiple facilities/locations for research activities.

b. **Project summary and field impact:** Provide an overview of your core research goal and the benefits you think it could provide to the broader CDR field. Tell us how your project meaningfully advances knowledge on at least one of the priority innovation areas identified in the RFP. What needs to happen for your project to unlock step-change advancements in cost and scale for this innovation area? Please include figures and system schematics, and be specific but concise. <1000 words

The current state of modeling for Measurement, Reporting and Verification

Monitoring, Reporting and Verification (MRV) for mCDR (defined here as any CDR pathway with an ocean outfall where the CO_2 removal from the atmosphere occurs in the ocean) requires complex ocean modeling informed by measurements to capture the mixing dynamics of plumes from a project in the nearfield, and to quantify CDR via air-sea CO_2 uptake that occurs broadly across space and time at regional and global scales. This motivates a need for accurate, cost-effective, operational ocean models both for CDR quantification at regional and larger scales, and to inform regulatory compliance, environmental impact assessments, and operational decision making. However, high computational expense and time-intensive expert development needs of legacy ocean models have created a bottleneck to achieving accurate, cheap, operational models for MRV. **This presents an opportunity to unlock a step-change advancement in lowering the cost of MRV, while improving accuracy, through leveraging innovative new modeling software and hardware technologies. This will enable more accurate carbon removal quantification, better uncertainty quantification, and effective sensor placement.**

Novel modeling software using modern software languages (Julia), together with advances in GPU chip infrastructure, have demonstrated a step-change in model performance (Fig. 2)[1,2]. This modeling framework also opens up a paradigm shift in approach to computational modeling for ocean physics and biogeochemistry - approaching computational modeling from an adaptable, hybrid approach that includes a process-based model foundation, with a framework to develop novel data-driven biogeochemical model representations. However, the existing open source projects that leverage the latest software and GPU-hardware infrastructure have, so far, largely been developed with idealized models or global, climate-scale model simulations in mind. This modeling software also has the potential to simulate coastal nearfield and regional processes required for MRV, but critical capabilities needed to accurately simulate coastal and regional processes are currently missing.

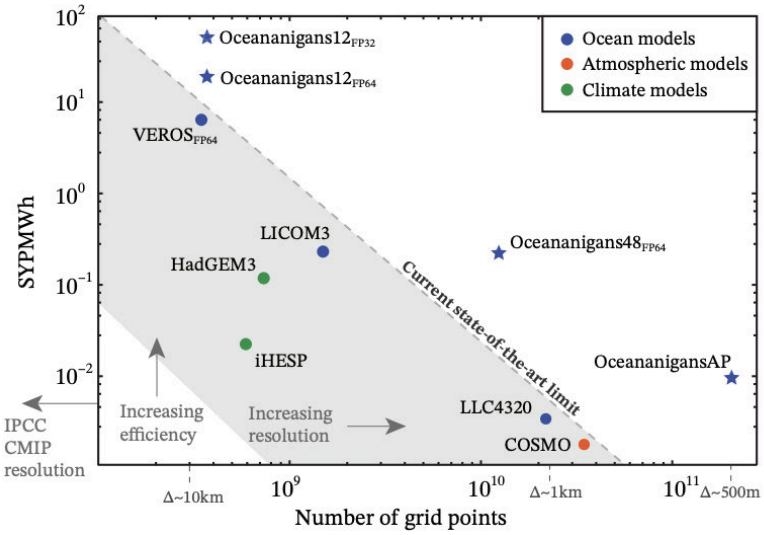


Figure 1 from Silvestri et al. (2023)[1]: Simulated years computed by a megawatt-hour of energy (SWPMWh) versus number of grid points for state-of-the-art atmosphere and ocean models. Stars show the performance of our ocean model in a realistic and “aqua planet” (AP) setup.

Modeling biogeochemical processes requires custom models that are able to represent different scales at different resolutions, and include detailed realistic representation of coastal features, river inputs, tides and wind forcing. While some legacy ocean model software libraries have these capabilities, they also have very high computational and development costs, and are not built to leverage the best available hardware (GPUs) and software (modern programming languages and hybrid AI workflows). We propose to develop the necessary underlying modeling capabilities required for accurate, economical, fit-for-purpose regional model configurations built upon the novel GPU-enabled software in NumericalEarth.jl[2,3], with underlying physics from Oceananigans.jl[2] and biogeochemistry from OceanBioME.jl[4]. Specifically, we will focus on developing the ability to 'nest' models - unlocking the ability to configure multiple models operating in tandem at resolutions appropriate for processes best modeled at different scales, and add differentiable programming - creating the foundation for robust uncertainty quantification, biogeochemical model optimization, and observing system design.

Model nesting

A key approach to addressing this challenge without ballooning the size and cost of a model is by running one or multiple higher resolution model grid(s) in a coastal area, inside of coarser resolution larger domains (i.e. nesting, Fig. 2). The state from the larger model grid can then communicate inflow and tracer information with the inner model grid (as in one way nesting, and vice versa (as in 2-way nesting). The key value of nesting is that different ocean processes operate at different scales, and modeling them accurately requires matching model resolution to process scale. Rather than running a single computationally prohibitive global high-resolution model, nesting enables seamless two-way data exchange between models at appropriate resolutions (from turbulent nearfield mixing to large-scale ocean circulation) making it practical to capture all processes relevant to mCDR quantification and field operations in a single framework. This approach is currently used in legacy ocean modeling systems for operational modeling and forecasting coastal systems (e.g. in Halifax Harbour[5]), but is not yet implemented in new, budget-friendly GPU-enabled modeling systems. **We propose to develop two-way nesting capability to enable multiple resolution, high fidelity regional simulations using modeling software, building upon the current NumericalEarth.jl software.**

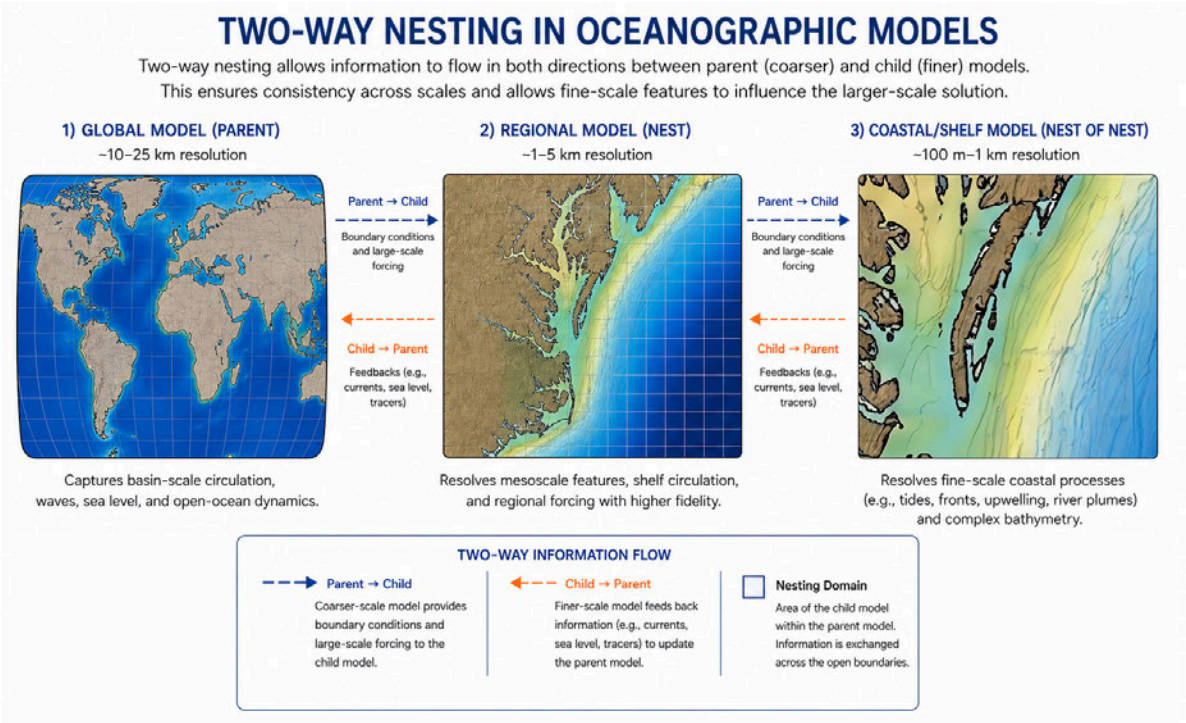


Figure 2: Illustration of nesting in oceanographic models.

Model differentiability

Ocean models rely on parameterizations for sub-grid processes that cannot be fully resolved. These parameters, along with poorly constrained boundary conditions, are dominant sources of model structural and parametric uncertainties. These uncertainties are currently first order uncertainties in CDR credit quantification for OAE projects. A powerful functionality that unlocks the ability to reduce uncertainties in CDR quantification and optimize project parameters and observing systems for project operations is model differentiability (Fig. 3). Differentiable programming and general-purpose Automatic Differentiation enables a range of derivative-based applications including model calibration, surrogate model training using machine learning, sensitivity analysis, uncertainty quantification, and quantitative observing system design. Developing a differentiable model also lays the groundwork for implementing variational data assimilation, which can further increase model skill by constraining the model biogeochemical observations (e.g. ECCO[6]). The DJ4Earth project[7] has developed open source Automatic Differentiation tools for Oceannanigans.jl, but the coupled biogeochemical models in OceanBioME.jl, which are essential for modeling mCDR processes, are not yet differentiable, limiting the ability to tune model parameterizations to measured data in biogeochemical models or optimize placement of biogeochemical sensors to minimize uncertainties through quantitative observing system design. **We propose to develop the differentiable biogeochemical models and software framework needed to unlock the ability to use data-driven methods to train biogeochemical models and optimize observation system design to reduce primary sources of uncertainty for mCDR quantification.**

These innovations will increase quantification fidelity by enabling models best suited for specific processes to operate in tandem, they will reduce computation cost through use of GPU architectures,

and they will directly enable measurement and reduction of uncertainty in models in a more cost-effective manner.

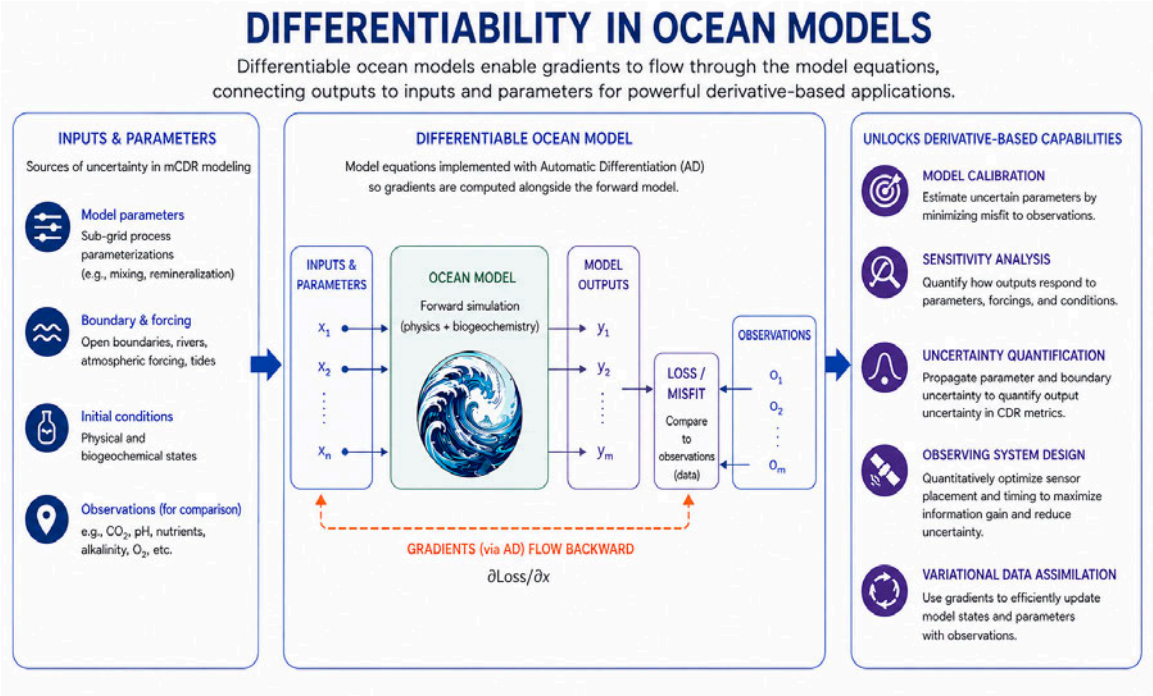


Figure 3: Illustration of differentiability in an ocean model and key derivative-based applications.

1. Silvestri, S., Wagner, G., Hill, C., Aradakani, M., R., Blaschke, J., Campin, J. M., ... & Ferrari, (2023). Oceananigans.jl: A model that achieves breakthrough resolution, memory and energy efficiency in global ocean simulations. *arXiv preprint*, arXIv, 2309
2. Wagner G.L., Silvestri, S., Constantinou, N. C., Ramadhan, A., Campin, J.M., Hill, C., Chor, T., Strong-Wright, J., Lee, X. K., Poulin, F., Souza, A., Burns, K. J., Bishnu, S., Marshall, J., Ferrari, R. (2025). *Submitted to Journal of Advances in Modeling Earth Systems*, arXiv:2502.14148
3. Wagner, G. L. et al (2026). NumericalEarth/NumericalEarth.jl: v0.1.1(v0.1.1). Zenodo. DOI: 10.5281/zenodo.18665405
4. Strong-Wright, J., Chen, S., Constantinou, N. C., Silvestri, S., Wagner, G. L., and Taylor, J. R. (2023). OceanBioME.jl: A flexible environment for modelling the coupled interactions between ocean biogeochemistry and physics, *Journal of Open Source Software, 90*(8), 5669. DOI: 10.21105/joss.05669
5. Laurent, A., Wang, B., Atamanchuk, D., Rakshit, S., Azetsu-Scott, K., Algar, C., & Fennel, K. (2025). A high-resolution nested model to study the effects of alkalinity additions in Halifax Harbour, a mid-latitude coastal fjord. *EGUsphere, 2025*, 1-37
6. Forget, G. A. E. L., Campin, J. M., Heimbach, P., Hill, C., Ponte, R. M., & Wunsch, C. (2015). ECCO version 4: An integrated framework for non-linear inverse modeling and global ocean state estimation. *Geoscientific Model Development, 8*(10), 3071-3104
7. Moses, W. S., Cheng, G., Churavy, V., Gelbrecht, M., Klöwer, M., Kump, J., Morlighem, M., Williamson, S., Apte, D., Berg, P., Giordano, M., Hill, C., Loose, N., Montoison, A., Narayanan, S. H. K., Pal, A., Schanen, M., Silvestri., S., Wagner., G. L., Heimbach, P. (2025) DJ4Earth: Differentiable, and performance-portable Earth System Modeling via program transformations, *ESS Open Archive* DOI: 10.22541/essoar.176314951.18114616/v1

c. **Project objectives:** Describe your project in more detail: what are your key research questions and proposed approach for answering them, including methods, project location, and scale where applicable? Tell us why your approach is best-in-class, and how you're differentiated from any other

organization working on a similar approach. Please include figures and system schematics, and be specific but concise. 1000-1500 words

Our primary research and development goal is to build the functionality needed to run GPU-enabled models at nearfield and regional scales and to quantify the improvements in model skill and reduction in uncertainty of CDR quantification compared to legacy ocean models. Within this overarching research goal there are two key open source model capabilities to be developed in parallel: nesting of model domains with increasing resolution, and leveraging differentiability to improve and optimize BGC models and reduce primary sources of uncertainty. Following the completion of development and initial testing for stability and performance, these two new capabilities will be implemented (both separately and together) in an existing regional model domain at one existing mCDR project site to assess the impact of these developments on model skill, cost, and uncertainty reduction.

GPU-native modeling capability 1: Nesting of model domains at multiple resolutions

The first model development track focuses on building and implementing the capability to nest one or more regional fine-resolution domains within a coarse-resolution domain using the NumericalEarth.jl framework, to cost-efficiently increase the fidelity and reduce uncertainty of modeled coastal processes. Our approach to the nesting development track is broken down into the following steps:

- **Testing of offline model nesting with existing model configurations**
  We will begin by implementing and testing offline nesting, in which a high-resolution regional model is forced using boundary conditions derived from a coarser-resolution parent simulation. This step allows us to validate consistency between parent and child domains, quantify boundary-driven errors and artifacts and establish baseline improvements in coastal process representation (e.g., circulation, stratification, tracer transport).

- **Online one-way regional model nesting capability development**
  In one-way nesting, the parent model drives the child model without feedback. This provides a computationally efficient pathway to significantly improve nearfield resolution while maintaining large-scale consistency. We will develop an online one-way nesting framework within the NumericalEarth.jl ecosystem, enabling dynamic coupling between parent and child domains during runtime, consistent exchange of velocity, tracer, and surface forcing fields and efficient GPU execution across nested domains

- **Exploration of two-way nesting capability**
  We will explore extending the framework to support *two-way nesting*, where the high-resolution child domain feeds information back to the parent model and fine-scale coastal processes influence larger-scale circulation and tracer distributions. This is particularly important for mCDR applications, where localized interventions can propagate beyond the nearfield and influence regional carbonate chemistry dynamics. We will evaluate numerical stability, conservation properties, and error propagation in this bidirectional coupling.

- **Implementation of nesting in an existing regional model configuration**
  We will apply the nesting framework to evolve an existing single grid regional model configuration of NumericalEarth.jl and OceanBioME.jl at an active or proposed mCDR field site (we currently have existing or in development regional configurations in the Salish Sea, the Mediterranean Sea, and Canary Current Upwelling System). We will configure nested

domains to resolve nearfield (<1 km) and regional (1–10 km) scales.

- **Verification of cost and uncertainty reduction and skill improvement for a regional model configuration**
  We will quantitatively evaluate the impact of nesting by comparing nested and non-nested configurations in terms of model skill (e.g., comparison to observations of physical and biogeochemical parameters), uncertainty reduction in key CDR metrics (e.g., air–sea CO_2 uptake) due to improvement in representation of small scale coastal processes, and computational cost efficiency, particularly leveraging GPU acceleration

GPU-native modeling capability 2: Differentiable framework for biogeochemical model optimization and uncertainty reduction

The second development track focuses on developing the differentiable version of the OceanBioME.jl software package. Once differentiability is formally achieved and the stability of the adjoint tested, we will leverage this differentiability to do adjoint sensitivity analysis, and develop a proof-of-concept framework for optimizing observing system design to minimize uncertainty in modeled CDR and other key biogeochemical parameters. The differentiable model development and application is broken down into the following steps:

- **Developing the differentiable version of OceanBiome.jl coupled to Oceananigans.jl**
  This involves ensuring all biogeochemical process formulations are compatible with automatic differentiation, removing or reformulating non-differentiable operations and validating consistency between forward and adjoint model behavior. This will build on the work of the DJ4Earth project[7], leveraging the Julia Enzyme.jl Automatic Differentiation tool.

- **Extending the control space (i.e., what parameters we are differentiating with respect to)**
  We will define and expand the control space for sensitivity and optimization, including, air-sea CO_2 flux, initial and boundary conditions for tracers (e.g., DIC, alkalinity, nutrients), and biogeochemical model parameters (e.g., reaction rates, biological uptake coefficients). This enables systematic exploration of which factors most strongly influence CDR outcomes and where uncertainties originate.

- **Testing the stability of the adjoint model**
  We will develop and rigorously test the adjoint (reverse-mode) model to ensure numerical stability over relevant simulation timescales, consistency between forward and adjoint sensitivities, and robustness to nonlinearities in coupled physical–biogeochemical processes

- **Configuring reference test cases for a small set of target quantities of interest**
  We will define a set of benchmark test cases focused on mCDR-relevant outputs, such as cumulative air-sea CO_2 uptake, changes in DIC and alkalinity inventories and spatial distribution of carbon sequestration. These test cases will serve as controlled environments to evaluate sensitivity structure, uncertainty propagation, and optimization performance.

- **Conduct convolution-based reconstruction experiments to determine sensitivity of air-sea CO_2 uptake and other mCDR-relevant parameters**
  Using an approach previously applied to understand controls on global productivity and air-sea carbon flux[8], we will perform convolution-based reconstructions to link perturbations to model responses. With these convolutions we will quantify the temporal and spatial

sensitivity of CO_2 uptake to forcing and parameter changes and identify dominant drivers of uncertainty in mCDR quantification. This provides a physically interpretable framework for understanding how interventions propagate through the system.

While nesting and differentiability exist in legacy modeling systems, implementing them in Oceananigans.jl unlocks something qualitatively new: a GPU-accelerated, differentiable, multi-scale platform purpose-built for mCDR. This combination enables cheaper and more accurate quantification through faster ensemble runs and reduced systemic bias; more powerful uncertainty quantification through the model's differentiability, enabling rapid sensitivity analysis and gradient-based optimization; and a practical pathway to cost-effective operational MRV, including dynamic sensor placement and dosing optimization, that legacy systems are too slow and inflexible to support.

Our team's direct contributions to Oceananigans.jl and OceanBioME.jl place us uniquely at the intersection of software development and mCDR application, allowing us to build these capabilities faster and with deeper fidelity than groups working at arm's length from the underlying tools. The Submarine team has a uniquely deep, practice-based perspective of the end-to-end challenges of modeling for MRV for mCDR. We have worked directly for suppliers to build fit-for-purpose modeling software, as a subject matter expert for the XPrize Carbon Removal mCDR finalist teams, as a verifier of the first OAE third-party verified carbon credits, and with academic and nonprofit mCDR researchers to develop protocols and tools to aid in data transparency and synthesis. By working directly with suppliers to develop fit-for-purpose models, we have gained firsthand practical insight into the biggest modeling bottlenecks and challenges facing this new sector.

8. Dutkiewicz, S., Follows, M. J., Heimback, P., & Marshall, J (2006). Controls on ocean productivity and air-sea carbon flux: An adjoint model sensitivity study. *Geophysical Research Letters, 33*(2).

d. **Risks:** What are the biggest risks to your project's success and how will you mitigate those? Include scientific; technical; project execution; measurement, reporting and verification (MRV); ecosystem; financial; and any other risks. <500 words

Risks to the project fall primarily in three categories: (1) dependence on third party platforms and components, (2) project complexity, and (3) inherent issues with ocean models.

The project depends on the open source NumericalEarth.jl platform (with Oceananigans.jl and OceanBioME.jl as the underlying physics and biogeochemical models), the leading GPU-native platform for realistic ocean physical and biogeochemical modeling. Oceananigans.jl support has broadened dramatically in recent years, but the platform is relatively new and less scientifically validated compared to established ocean modeling platforms such as ROMS and MITgcm, even though they are slower and much less versatile. Additionally, the project depends on other open source initiatives feeding into the platform, in particular Automatic Differentiation, which is currently under development by the DJ4Earth project. Submarine will leverage strong existing connections with the core developers of NumericalEarth.jl to coordinate on parallel development efforts and targeted validation studies. Submarine will additionally mitigate these risks through contribution and leadership, which will benefit the platforms both directly, and indirectly through enhancing the

perception that the platforms are a thriving ecosystem.

The proposed project is complex - developing a differentiable version of OceanBioME.jl involves complex mathematics; nesting involves complex interactions between software components which are themselves inherently complex with many subsystems. Any complex project necessarily involves risk, and given the deep domain knowledge required, it limits the available pool of expertise capable of tackling these challenges. Submarine will mitigate these risks partly through careful management, partly through our access to internal and external expertise for project work, and partly through involving external domain experts as reviewers and advisors.

Finally, all ocean models today do not fully represent all known ocean processes, and relevant ocean processes may exist that are not yet known or understood. This is partly an inherent limitation of 'modeling', and partly a reflection of the state of ocean science. As a result, all ocean models make slightly different decisions about what and how to model, with the result that models can exhibit systemic biases of various kinds. The primary mechanisms for reducing these uncertainties are higher fidelity sensor data which can be assimilated into models to help them match real world observations, field research to uncover and quantify relevant ocean processes, and inter-model comparison to identify sources of bias. Submarine is actively involved with community efforts in all three of these approaches. We note that this risk exists in all modeling of poorly constrained complex systems and is not unique to this project.

e. **Proposal to Frontier:** Please list your primary research deliverables, project timeline, and requested funding below. If you are selected for Frontier funding, this table will form the basis of contract discussions.

Primary research deliverables *(List tangible outputs of the project: e.g., validated datasets, peer-reviewed models, bench-scale prototypes, or field trial results. Should focus on resolving a specific bottleneck or uncertainty identified in Section 1.a)*	Open source development of key capabilities to enable regional, fit-for-purpose for mCDR MRV ocean modeling. This includes development and application of 1) regional nested model domains and 2) leveraging differentiability of GPU-enabled modeling software to optimize biogeochemical models and reduce uncertainty, expanding on the existing NumericalEarth.jl and OceanBioME.jl packages.
Field impact *(Briefly describe how these deliverables resolve key bottlenecks for a specific CDR pathway or broader CDR field, e.g. efficiency gains, scaling unlocks, or reducing measurement uncertainty.)*	The completion of the proposed work will lead to a step change reduction in computational costs of modeling for MRV, while improving quantification accuracy and improving uncertainty quantification and reduction. These deliverables are expected to significantly reduce the cost and

	time-intensiveness of MRV for open system pathways that are reliant on numerical modeling, while simultaneously creating a framework to accelerate data-driven improvements in ocean biogeochemical models, reducing a primary uncertainty in CDR quantification for mCDR pathways.
Project timeline & key milestones *(At what point should Frontier consider the research project complete? Include major progress markers. Frontier generally prefers projects with clear readouts within ~18 months.)*	Completion of offline model nesting testing: October 2026 Completion of formal differentiability of OceanBioME.jl: April 2027 Completion of online regional model nesting development: June 2027 Completion of investigation of two-way nesting: December 2027 Conduct convolution-based reconstruction experiments to determine sensitivity of air-sea CO2 uptake and other mCDR-relevant parameters: April 2028 Implementation of nesting to existing regional model configurations to evaluate of uncertainty reduction and skill improvement: July 2028 Overall Project completion: July, 2028
Total grant request (US$) *(The total funding required to execute the research scope described above. Frontier R&D grants typically range from US$250k–$750k.)*	$426,000